PDF OPERATION MANUAL

VCA Synthetic Research Workbench

How a Visa/VCA reviewer can upload a real survey PDF, run Swiss synthetic persona agents with IBM watsonx.ai / Granite, and interpret aggregate insights, persona-level responses, validation checks and a consultant decision brief.

What this proves

- Flexible survey input: no fixed question set.
- PDF upload and extraction audit for realistic partner testing.
- Swiss public-data-grounded persona simulation.
- Consultant-ready aggregation plus individual response traceability.
- Benchmark, consistency, coverage and realism validation.
- Real-model proof path using IBM watsonx.ai / Granite 4 H Small.

Demo artifacts

Application:
https://visa-synthetic-research-copilot.27cqtktlikeo.eu-de.codeengine.appdomain.cloud

Repository:
https://github.com/ZihaoLu001/visa-synthetic-research-copilot

Sample PDF:
Federal Reserve mobile-payments survey excerpt adapted for Visa demo use.

Live model:
IBM watsonx.ai in eu-de, default model ibm/granite-4-h-small.

Public-data prototype. No Visa internal or client-sensitive data is used.

START HERE

Reviewer Workflow At A Glance

Who should use this?

Visa Consulting & Analytics reviewers, IBM mentors, or student evaluators who want to test the prototype with a survey/interview artifact instead of a hardcoded prompt.

12
Default live Granite proof respondents

96
Full presentation panel available

PDF
Recommended partner upload format

4
Validation layers in the cockpit

POSITIONING

This is an early-stage hypothesis screening tool. It helps consultants identify promising propositions and survey design risks before commissioning real customer research.

Four-step operation

1 UPLOAD
Attach a PDF survey or use the sample public survey.

2 REVIEW
Check extracted survey text and edit questions if needed.

3 CONFIGURE
Set target market and adjust card propositions.

4 RUN
Generate insights, persona responses and validation evidence.

RECOMMENDED REVIEWER SETTING

Use **Model provider = watsonx** for the final real-model proof. Keep **mock** only as a fallback for rehearsal or quota issues. Use **Quick real-model proof** for a fast live run, then move to **Full survey** and **96 respondents** when quota/time allows.

NOT A TOY DEMO

Model, Algorithm And Delivery Stack

Real LLM path

IBM watsonx.ai is the primary provider. The app calls Granite through the ibm-watsonx-ai ModelInference abstraction, defaulting to ibm/granite-4-h-small in eu-de.

Fallback path

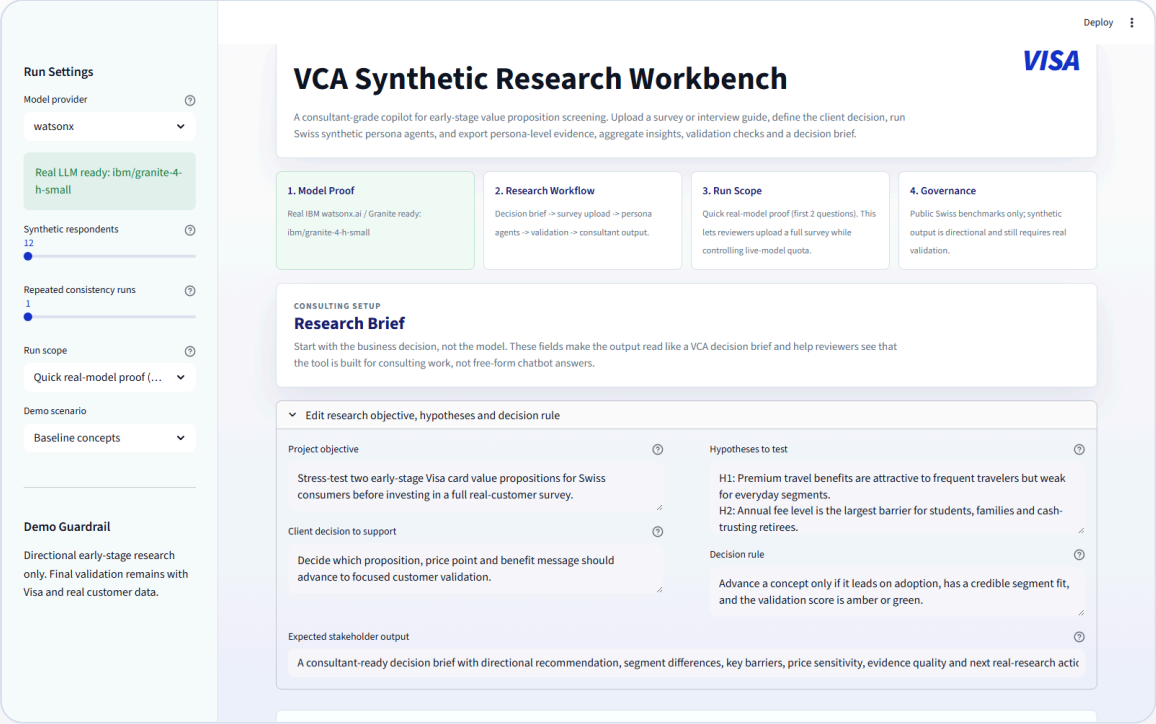
MockLLM is deterministic and exists only for CI, rehearsal and classroom quota contingency. It is not presented as real customer intelligence.

Core algorithm

Extract survey text, parse structured questions, sample Swiss micro-personas, run one respondent agent per persona, aggregate weighted results, validate, and synthesize a VCA decision brief.

Consulting guardrail

Synthetic results are directional. They help screen concepts and design better real customer research; they do not replace final Visa validation.



INPUT ARTIFACT

Step 1 - Upload A Real Survey PDF

Preferred format

Use PDF for partner review because it matches how survey decks and interview guides are usually shared.

Sample file

The right panel lets reviewers download the included public PDF survey excerpt and upload it immediately.

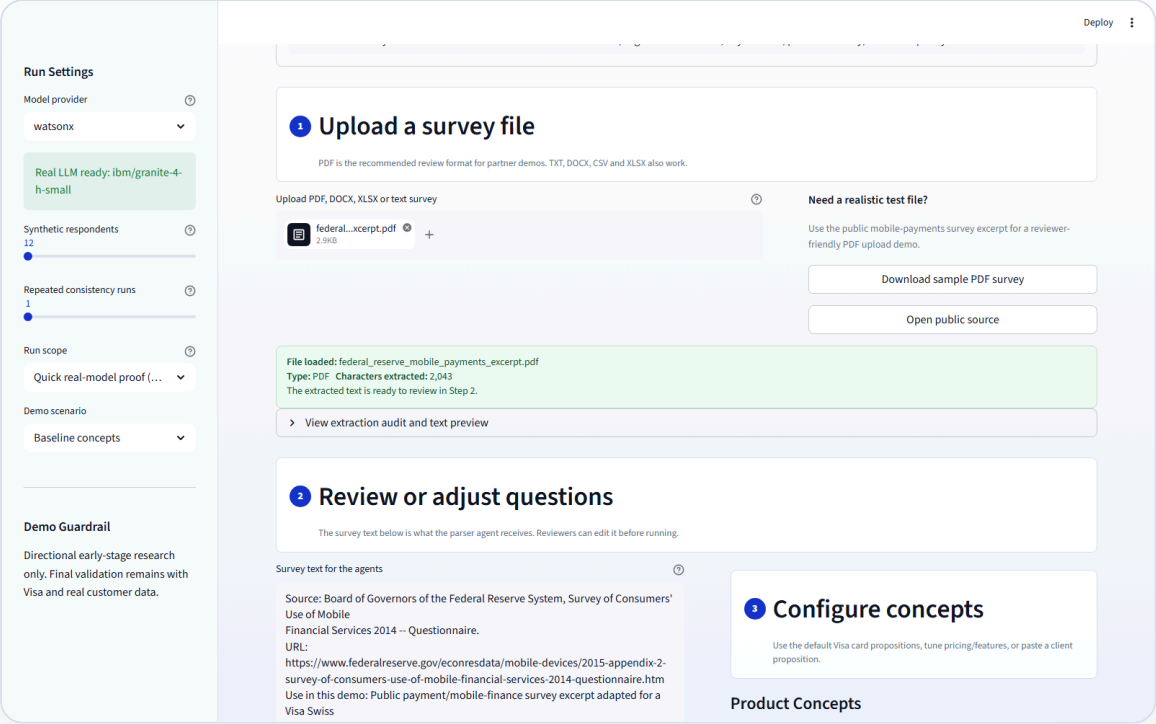
Extraction proof

After upload, the app shows file name, file type and extracted character count before any agent run.

Data guardrail

The sample comes from a public Federal Reserve questionnaire; no Visa internal data is required.

Public source: https://www.federalreserve.gov/econresdata/mobile-devices/2015-appendix-2-survey-of-consumers-use-of-mobile-financial-services-2014-questionnaire.htm



SURVEY PARSER INPUT

Step 2 - Review Or Adjust Questions

Editable extracted text

The textarea is the exact survey text sent to the parser agent. Reviewers can leave it unchanged or edit wording.

Why this matters

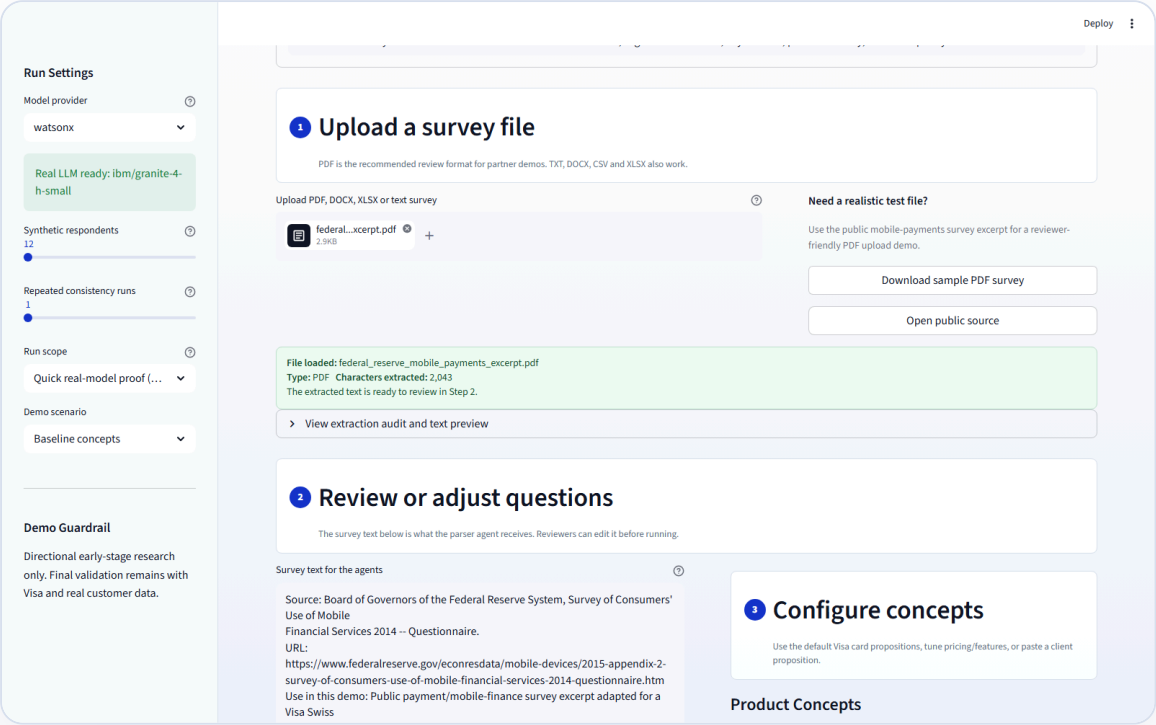
This directly addresses the Visa clarification that the system should answer survey questions without a fixed limitation.

Supported artifacts

The same ingestion layer accepts PDF, DOCX, XLSX, CSV, TXT and Markdown.

Audit trail

If the extracted text is edited, the run records that fact in the Question Parser tab.



AGENT WORKFLOW

Step 3 - Configure Concepts And Run

Concept testing

Use the default Premium Travel Card and Everyday Cashback Card, or paste a new proposition.

Live sensitivity

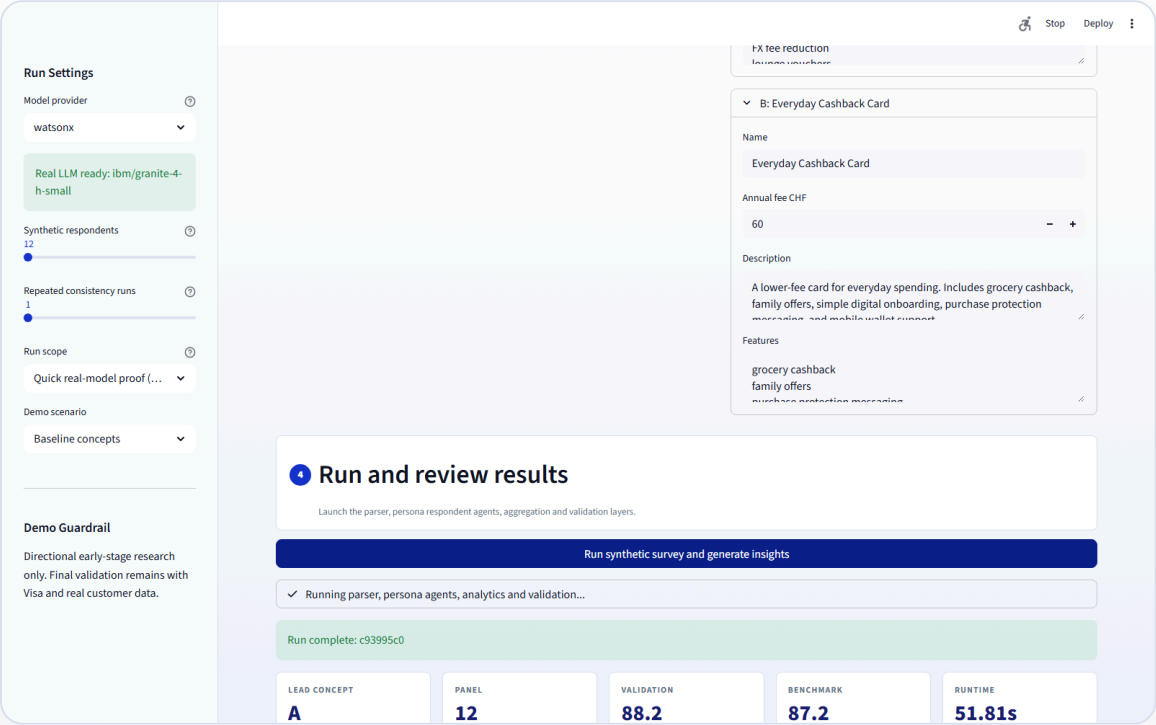
Change the annual fee or protection messaging, rerun, and compare movement in segment fit and adoption index.

Panel scale

The quick watsonx proof starts small to conserve quota. The full 96-persona run expands seven Swiss archetypes into a weighted micro-population.

Run button

The button launches survey parsing, persona respondent agents, analytics, validation and decision-brief synthesis in one workflow.



AGGREGATED OUTPUT

Step 4 - Read Consultant-Level Insights

Recommendation

The first tab summarizes the leading concept and the reason behind the result.

Adoption index

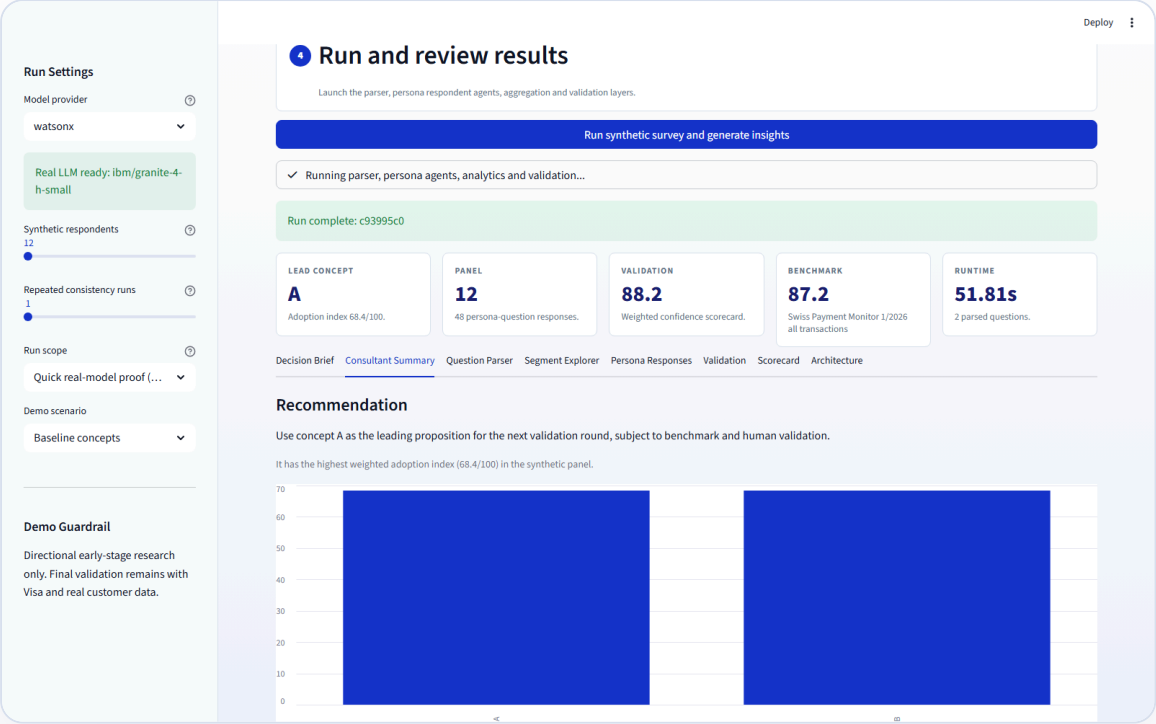
The bar chart compares concept fit across the synthetic Swiss panel.

Pricing signal

Acceptable fee ranges help frame a real customer price-sensitivity test.

Feature and barrier signals

Top labels surface recurring needs, trust drivers and objections.



QUESTION PARSER TAB

Confirm The Survey Was Parsed From PDF

Input Source Audit

Shows Source = uploaded_file, File type = pdf, character count and extraction notes.

No hardcoding

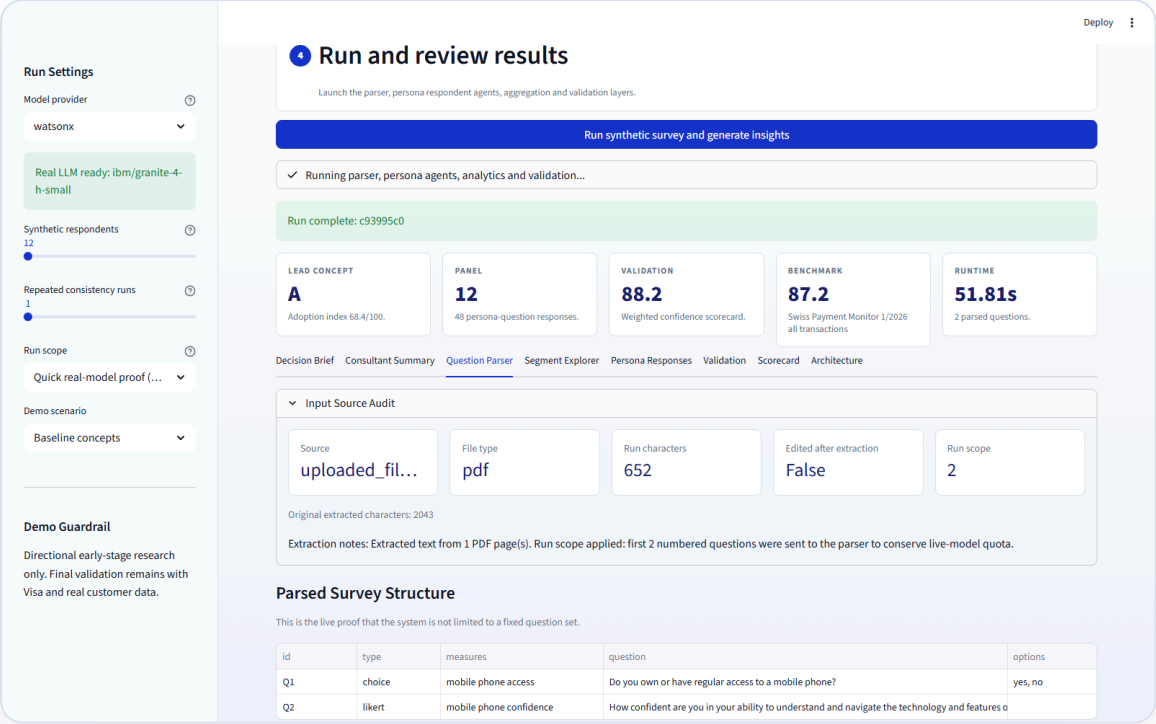
The parsed table lists question id, type, construct and options generated from the uploaded survey.

Demo expectation

Open this tab during review to prove the app can accept a new survey artifact.

Quality fix

The parser handles PDF line wrapping so wrapped questions stay together.



SEGMENT EXPLORER TAB

Inspect Segment Differences

Why segments matter

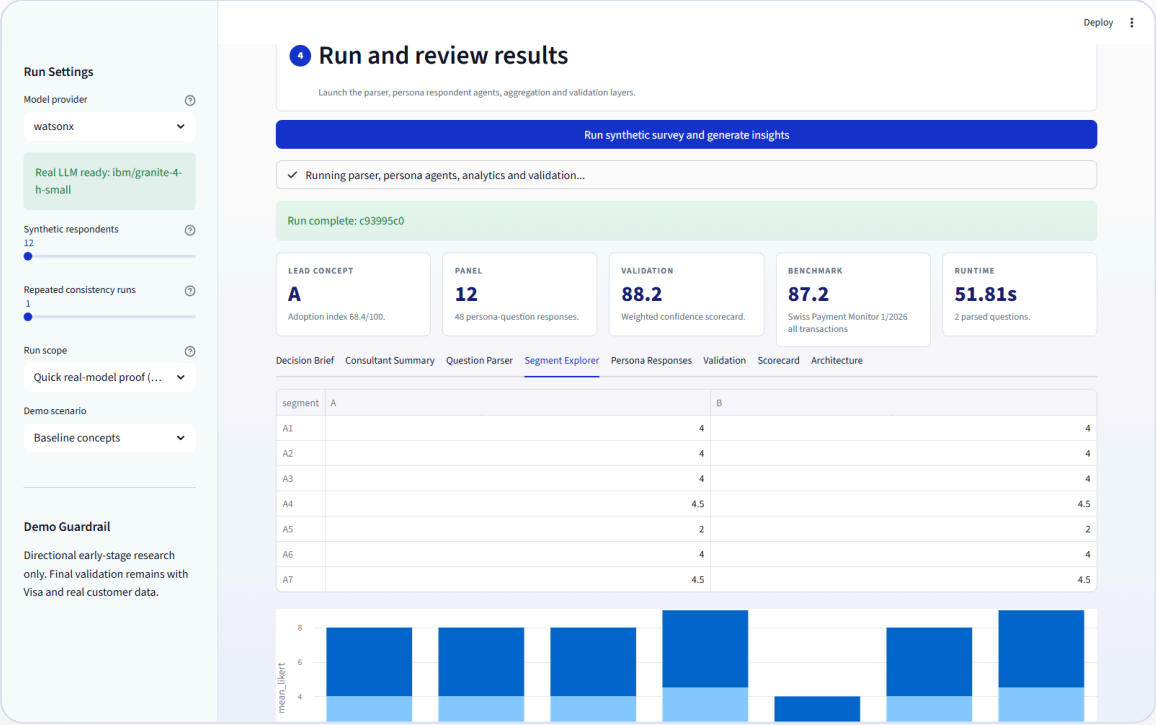
Visa asked for different personas or demographic groups, not one generic customer.

What to look for

Compare how students, family budget parents, travelers, retirees and SME owners respond differently.

Consultant use

Segment variation helps decide which customer group deserves real follow-up research.



PERSONA RESPONSES TAB

Inspect Persona-Level Traceability

Individual responses

Every row stores persona id, persona name, concept, question, answer, rationale and confidence.

Filter workflow

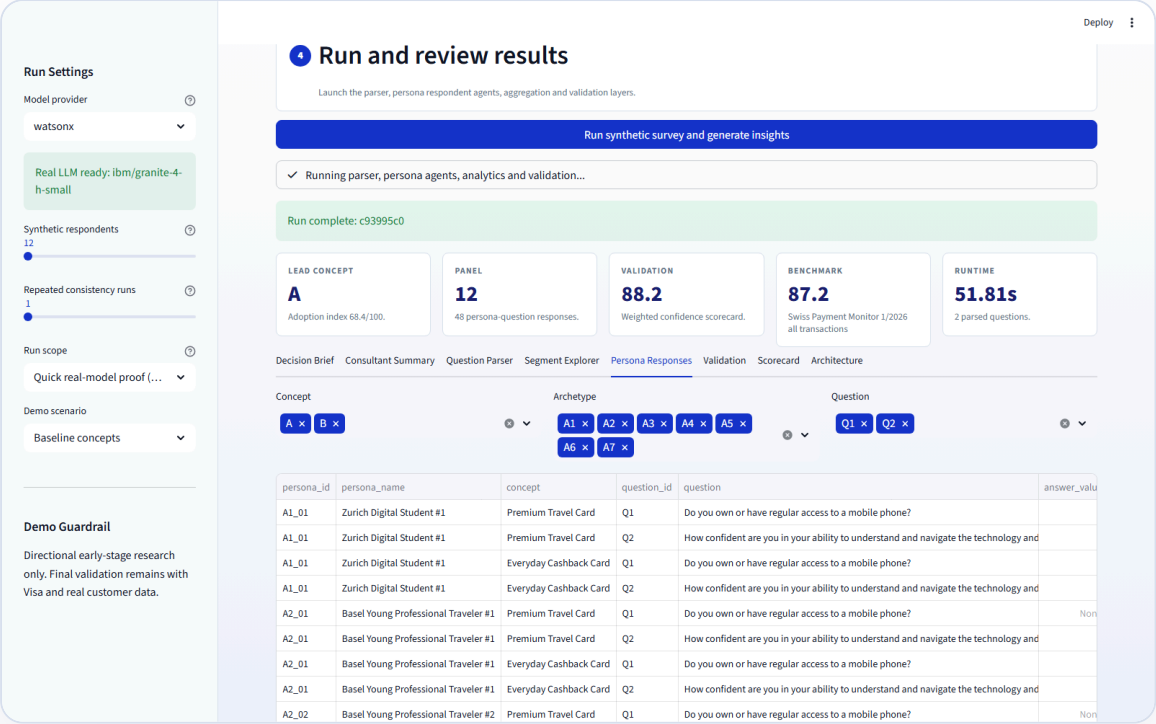
Filter by concept, archetype or question to inspect why a segment is driving the aggregate result.

Reviewer value

This satisfies the Visa request for both aggregated insights and persona-level survey responses.

Export

Download the filtered response table as CSV for analysis or appendix material.



VALIDATION TAB

Check Validation Confidence

Benchmark alignment

Compares synthetic payment behavior mix against public Swiss payment survey anchors.

Internal consistency

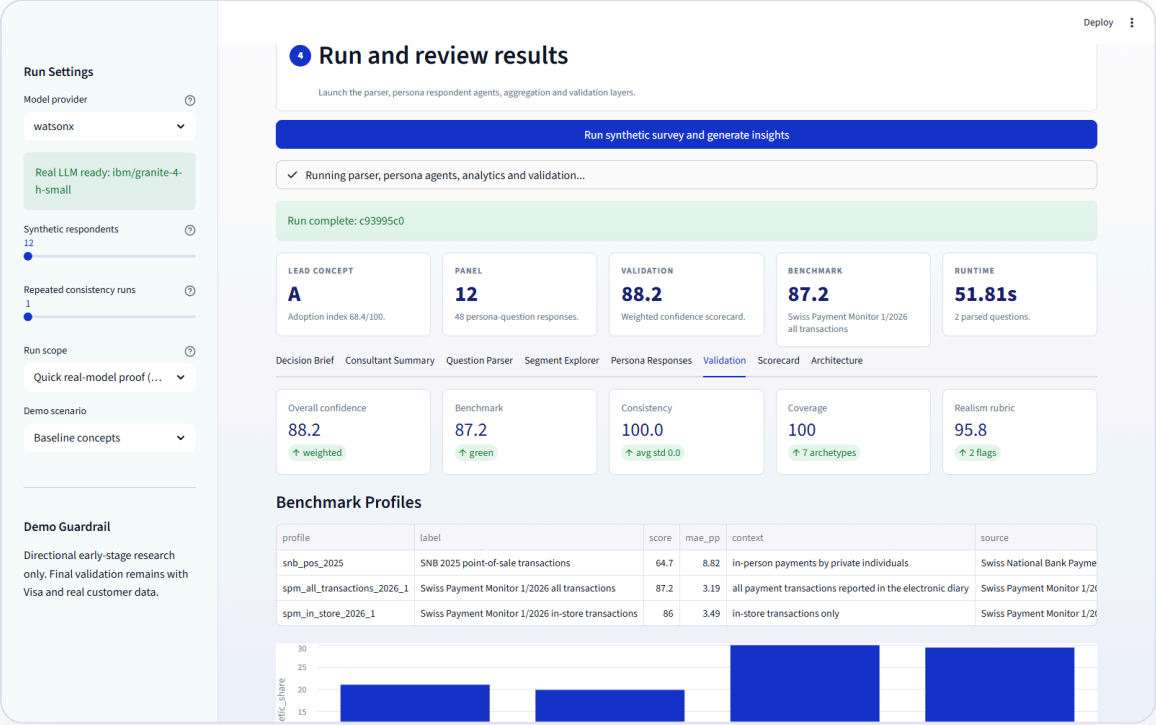
Checks repeated persona runs for response variance.

Coverage

Confirms the panel covers the Swiss archetypes and the survey covers key constructs.

Interpretation

This is directional evidence, not a claim that synthetic respondents replace real Visa validation.



EVALUATION MAPPING

Use The Scorecard In The Final Presentation

Demo

Running app, PDF upload, parser, agents, results and validation are all live.

Architecture

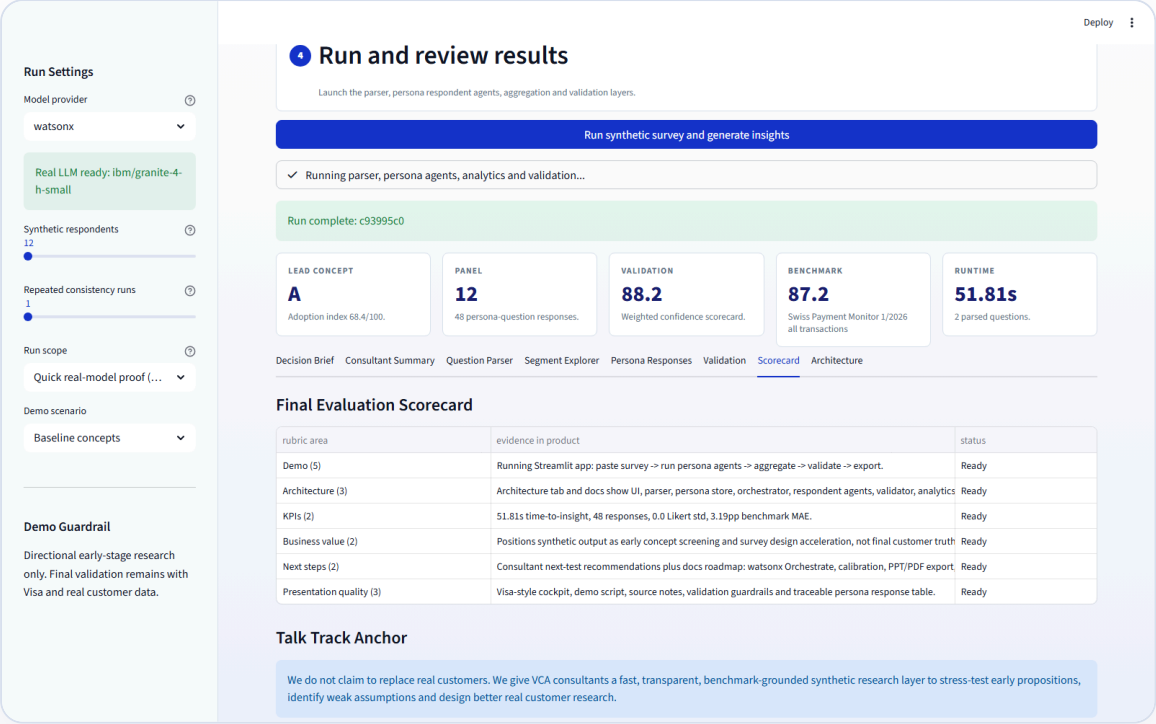
UI, ingestion, parser, persona store, orchestrator, respondent agents, analytics and validator are visible.

KPIs

Runtime, response count, parse success, consistency and benchmark scores are quantified.

Business value

Position the system as early-stage concept screening and better real-survey design.



PARTNER ALIGNMENT

Suggested Slack Message To Visa / IBM

Message draft

Hi Visa / IBM team,

We prepared a short operation manual for the current prototype. It shows a real PDF survey upload, dynamic parsing, IBM watsonx/Granite model path, Swiss synthetic persona responses, aggregated consultant insights, and validation checks.

Could you please let us know whether this matches your expected direction for the final Visa use case?

1. Is the flexible PDF survey/interview upload flow aligned with your expectation?
2. Should the final output emphasize aggregate insights, persona-level responses, or both?
3. Are the benchmark, consistency, coverage and realism validation checks useful enough for VCA consultants?
4. Is there any missing use-case requirement we should address before finals?

Thank you!

What to emphasize verbally

- **Flexible input:** The system is not a fixed Q&A bot.
- **Public-data grounding:** Personas are used because internal Visa customer data is unavailable.
- **Consultant value:** It produces early directional evidence and better real survey design.
- **Validation discipline:** The app exposes limitations through benchmark and consistency checks.

BEST CHANNEL

Use the Visa-specific Slack channel for use-case/product feedback. Use the general guest channel for deployment or IBM platform issues.